

Unit 4: Data Preprocessing

Data Reduction Techniques – Class Activity Sheet #8

Feature selection Dimensionality reduction Sampling and aggregation

Name		Batch	
Date		Global ID	

Learning Objectives

- Explain why reducing the size of a dataset can improve efficiency and interpretation.
- Distinguish feature selection from feature extraction.
- Apply variance and correlation based feature selection.
- Use SelectKBest and Principal Component Analysis for dimensionality reduction.
- Reduce the number of records through sampling and aggregation.
- Implement basic data reduction techniques using Python.

Part A Concept Recall

Data reduction decreases the number of rows, columns, or stored values while retaining the information needed for analysis. A smaller dataset can require less memory, train faster, and be easier to understand.

Data reduction branch	What is reduced	Techniques in this activity
Dimensionality reduction	Columns or dimensions	Feature Selection, Feature Extraction-PCA, LDA
Instance or numerosity reduction	Rows, representatives, or distinct values	Sampling, clustering, binning or discretization, and outlier removal

Part B Identify the Suitable Technique

1. A dataset contains 50 features, but only 8 are relevant to the target. Technique: _____
2. A column has the same value in every row. Technique: _____
3. Ten correlated sensor readings must be represented by two new components. Technique: _____
4. A dataset contains ten million records and a representative subset is sufficient. Technique: _____
5. Daily sales records are converted into monthly sales totals. Technique: _____

Part C Feature Selection

Feature selection keeps useful original columns and removes irrelevant, redundant, or low-information columns. The meaning and units of the retained features do not change.

Method	Simple explanation	Main use
Domain-based selection	Select or remove features using knowledge of the problem	Remove irrelevant identifiers and data-leakage features
Variance threshold	Remove features whose values change very little	Eliminate constant or nearly constant features
Correlation filter	Remove one of two features that contain almost the same information	Reduce redundancy and multicollinearity
SelectKBest	Statistically score every feature against the target and retain the best k features	Automatically select features that are strongly related to the target

Activity C1 Domain Based Selection

Student ID	Hours Studied	Attendance	Previous Marks	Final Marks
S101	2	65	55	58
S102	4	75	62	68
S103	6	85	72	80
S104	8	92	84	91

1. Identify the target column when predicting Final Marks.
2. Which identifier should normally be removed before model training?
3. List the possible predictor features.
4. Explain why Student ID is unlikely to describe academic performance.

Activity C2 Variance Threshold

Dataset: Classroom = [1, 1, 1, 1, 1] and Practice Tests = [1, 2, 3, 4, 5]

$$\text{Variance} = \frac{\sum (x_i - \bar{x})^2}{n}$$

A constant feature has variance 0 and cannot distinguish one observation from another.

5. Find the mean and variance of the Classroom feature.
6. Which feature should be removed using a variance threshold of 0?
7. Why does a zero-variance feature provide no useful separation between students?

```
import pandas as pd
from sklearn.feature_selection import VarianceThreshold
df = pd.DataFrame({
    "Classroom": [1, 1, 1, 1, 1],
    "Practice_Tests": [1, 2, 3, 4, 5],
    "Attendance": [70, 75, 80, 85, 90]
})
selector = VarianceThreshold(threshold=0)
reduced_array = selector.fit_transform(df)
selected_columns = df.columns[selector.get_support()]
reduced_df = pd.DataFrame(reduced_array, columns=selected_columns)
print(selected_columns)
print(reduced_df)
```

Activity C3 Correlation Based Reduction

Student	Height cm	Height m	Weight kg	Fitness score
A	150	1.50	48	62
B	160	1.60	55	70
C	170	1.70	63	78
D	180	1.80	72	86

Student	Height cm	Height m	Weight kg	Fitness score
E	190	1.90	82	94

Height cm and Height m carry the same information in different units. Their Pearson correlation is 1.

8. Which two predictor columns are perfectly correlated?
9. If the correlation threshold is 0.90, which one of these columns could be removed?
10. Does a high correlation automatically mean both features should be removed? Explain.
11. Why should the target column be excluded when searching for redundant predictors?

```
numeric_features = df[["Height_cm", "Height_m", "Weight_kg"]]
corr_matrix = numeric_features.corr().abs()
print(corr_matrix)
# Remove one feature from the highly correlated pair
reduced_df = df.drop(columns=["Height_m"])
print(reduced_df)
```

Activity C4 SelectKBest

SelectKBest assigns a statistical score to each predictor in relation to the target and retains the highest-scoring k features. The scoring function must match the learning problem.

Problem	Example target	Suitable score
Classification	Pass or Fail	f_classif
Regression	Final marks	f_regression

```
from sklearn.feature_selection import SelectKBest, f_regression
X = df[["Hours_Studied", "Attendance", "Previous_Marks"]]
y = df["Final_Marks"]
selector = SelectKBest(score_func=f_regression, k=2)
X_reduced = selector.fit_transform(X, y)
selected_features = X.columns[selector.get_support()]
print(selected_features)
print(selector.scores_)
```

12. What does k = 2 mean?
13. Why is f_regression used when Final Marks is numerical?
14. Does SelectKBest create new features or retain original features?

Part D Feature Extraction Using PCA

Principal Component Analysis creates new features called principal components. Each component is a weighted combination of the original numerical features. The first component captures the greatest possible variance, and later components capture the remaining variance while remaining uncorrelated with earlier components.

Term	Meaning
Principal component	A new feature formed from weighted original features

Term	Meaning
Explained variance ratio	Fraction of total variance represented by a component
n components	Number of principal components retained
Standardization	<i>Places features on comparable scales before PCA</i>

Activity D1 PCA Interpretation

Suppose PCA produces explained variance ratios $PC1 = 0.70$, $PC2 = 0.20$, and $PC3 = 0.10$.

15. What percentage of total variance is explained by PC1?
16. How much variance is retained by keeping PC1 and PC2?
17. How much variance is lost if PC3 is removed?
18. How many components should be retained to preserve at least 85 percent of the variance?

Activity D2 PCA with Python

```
import pandas as pd
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
df = pd.DataFrame({
    "Maths": [70, 60, 90, 55, 82, 68],
    "Science": [80, 65, 85, 60, 88, 72],
    "English": [75, 70, 95, 58, 84, 74]
})
X_scaled = StandardScaler().fit_transform(df)
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)
pca_df = pd.DataFrame(X_pca, columns=["PC1", "PC2"])
print(pca_df)
print("Explained variance ratio:", pca.explained_variance_ratio_)
print("Total retained:", pca.explained_variance_ratio_.sum())
```

19. How many original features are supplied to PCA?
20. How many new features are created?
21. Why are the features standardized before applying PCA?

Feature Selection and PCA Compared

Feature selection	PCA feature extraction
Keeps a subset of original features	Creates new principal components
Feature names and meaning remain available	Components can be harder to interpret
Useful when interpretability is important	Useful when many predictors are correlated

PART E INSTANCE/ROW REDUCTION

Activity E1 Sampling

Sampling selects a smaller group of rows. The sample should represent the larger dataset. Set `random_state` to obtain the same sample whenever the code is repeated.

```
# Select 30 percent of the rows
sample_df = df.sample(frac=0.30, random_state=42)
# Select exactly 100 rows
sample_100 = df.sample(n=100, random_state=42)
```

23. What fraction of rows is selected when `frac = 0.30`?
24. Why is `random_state = 42` used?
25. What is one possible risk of using a poorly selected sample?

Activity E2 Aggregation

Date	City	Sales
2026-01-01	Delhi	1200
2026-01-02	Delhi	1500
2026-01-01	Mumbai	1800
2026-01-02	Mumbai	1600

26. Calculate the total sales for Delhi.
27. Calculate the total sales for Mumbai.
28. How many city-level rows remain after aggregation?
29. What detail is lost when individual dates are replaced by city totals?

```
sales_df = pd.DataFrame({
    "City": ["Delhi", "Delhi", "Mumbai", "Mumbai"],
    "Sales": [1200, 1500, 1800, 1600]
})
city_total = sales_df.groupby("City", as_index=False)["Sales"].sum()
print(city_total)
```

Activity E3 Data Binning or Discretization

Binning replaces many continuous values with a smaller set of interval labels. It reduces the number of distinct values and may simplify patterns, but it also removes exact-value detail. Bins may use equal widths, equal frequencies, or boundaries chosen through domain knowledge.

Age	Domain based age group
18	18 to 25
23	18 to 25
29	26 to 35
34	26 to 35
42	36 to 50

Age	Domain based age group
57	Above 50

```
import pandas as pd
df = pd.DataFrame({"Age": [18, 23, 29, 34, 42, 57]})
df["Age_Group"] = pd.cut(
    df["Age"],
    bins=[0, 25, 35, 50, float("inf")],
    labels=["18 to 25", "26 to 35", "36 to 50", "Above 50"],
    include_lowest=True
)
print(df)
```

Part G Combined Practice Dataset

Goal: Reduce the employee dataset before building a model to predict Performance Score.

Employee ID	Age	Experience	Training Hours	Projects	Constant	Performance Score
E101	24	2	15	4	1	58
E102	30	6	25	8	1	74
E103	27	4	20	6	1	67
E104	35	10	30	12	1	90
E105	29	5	18	7	1	70
E106	32	8	28	10	1	84

Tasks - Write the Code

1. Create the dataset as a Pandas DataFrame.
2. Display the shape, column names, data types, and summary statistics.
3. Separate the predictors X and target y, where Performance Score is the target.
4. Remove Employee ID because it is an identifier.
5. Use VarianceThreshold to remove the Constant feature.
6. Create and display the correlation matrix for the remaining numerical predictors.
7. If any pair of predictors has absolute correlation greater than 0.90, remove one predictor from that pair and justify the choice.
8. Use SelectKBest with f_regression to retain the best two predictors.
9. Standardize the selected features and reduce them to one principal component using PCA.
10. Display the original dimensions and the reduced dimensions.

Starter Code

```
import pandas as pd
from sklearn.feature_selection import VarianceThreshold, SelectKBest, f_regression
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
data = {
    "Employee_ID": ["E101", "E102", "E103", "E104", "E105", "E106"],
    "Age": [24, 30, 27, 35, 29, 32],
    "Experience": [2, 6, 4, 10, 5, 8],
    "Training_Hours": [15, 25, 20, 30, 18, 28],
    "Projects": [4, 8, 6, 12, 7, 10],
    "Constant": [1, 1, 1, 1, 1, 1],
    "Performance_Score": [58, 74, 67, 90, 70, 84]
}
df = pd.DataFrame(data)
# Continue the solution below
```

Analysis Questions

1. How many rows and columns are present in the original dataset?
2. Which column is the target?
3. Which column is removed by VarianceThreshold and why?
4. What is the difference between removing Employee ID and applying PCA?
5. Why must PCA be fitted only on predictor features and not on the target?
6. What is one benefit and one possible disadvantage of reducing the number of features?